

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st November 2019

**Subject SP7 – General Insurance Reserving and
Capital Modelling**

**Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)
Total Marks: 100**

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)**
- i)** Outline the key aspects which should be considered while setting the expected loss ratio assumption in the expected loss ratio method. (2)
 - ii)** Define additional unexpired risk reserve (AURR). (1)
 - iii)** A student actuary is estimating IBNR reserves using the expected loss ratio method where he uses some ULR benchmarked to relevant portfolios relying on his experience and judgement. He also uses the same ULR for estimating the AURR. Explain the issues arising out of this calculation, if any, to be advised to the student actuary. (3)
 - iv)** Explain how the AURR reserves should be profiled or allocated in various duration buckets for the purpose of asset liability modelling (ALM). (5)
 - v)** For ALM in General Insurance, which one in your opinion would be more relevant cashflow matching or duration matching and why? (3)
- [14]**
- Q. 2)**
- i)**
 - a)** Define the three components of error in any stochastic reserving exercise. (3)
 - b)** Map the following into the three components of error stated in (a) above
 - System unable to capture the date on which reserves are revised
 - Change in legal/claims environment
 - Unusual large claims
 - Emergence of latent claims
 - Change in claims case reserving
 - Nature of loss data missing in some claims (3)
 - ii)** In practice it is reasonable to expect that the stochastic best estimate should be aligned to the actuarial best estimate as the stochastic best estimate should represent the uncertainty around the actuarial best estimate.
 - a)** Identify the stochastic methods which will meet the expectation mentioned above if the chain ladder method is strictly followed without any further adjustment. (2)
 - b)** In practice the two estimates, i.e. the Actuarial Best Estimate and the Stochastic Best Estimates are aligned by shifting the reserve distribution using either an additive or multiplicative scaling. In additive scaling, the standard deviation is kept the same whereas in multiplicative scaling (stretching or shrinking) the Coefficient of Variance is kept the same after scaling.

Suppose the chain ladder estimate is followed with some manual adjustment for estimating the actuarial best estimate and it is found that the actuarial best estimate thus derived is lower than the stochastic best estimate. Discuss the approach which you would advise in this case in order to align the two estimates, justifying with appropriate reasons. (2)

- iii)** A student actuary wants to estimate the reserving variability using some stochastic method on net claims data. Identify the issues with this approach. (3)
[13]

Q. 3) i) Define Override Commission and Ceding Commission. (2)

- ii)** As part of a recently introduced Sandbox approach by the regulator of a country interested in promoting innovation, a new InsureTech player is planning to introduce Usage based motor insurance. The device costs measuring the usage and behaviour are substantially high compared to the pure Own Damage Premium.
- a)** Outline the possible reinsurance arrangements which can help minimize these challenges for the insurer. (3)
- b)** Explain the challenges, if any, in the reinsurance arrangements and discuss the possible ways to overcome these challenges. (3)

- iii)** As the consulting actuary of the new InsureTech player, the senior management team has sought your advice on the amount of capital required to support the business. Outline your considerations in providing advice to them. (10)

- iv)** An actuarial analyst in the team has built a capital model. You have reviewed the model and noticed that there is no adjustment for catastrophe claims and large claims in the model.
- a)** Explain why the actuarial analyst needs to adjust claims data for catastrophe/large claims for the purpose of capital modelling? (8)

- b)** Explain different ways of modelling catastrophe claims. (5)
[31]

Q. 4) i) An investment banker helping few private equity investors interested in investing in existing general insurance companies, identifies few performance measurement parameters for valuation and comparison, among which are Combined Operating Ratio and Profit Margin.

A student actuary indicates that both parameters indicate the same profitability. Identify any issues with the statement made by the student actuary while defining these ratios. (3)

- ii)** One of the private equity investors is looking at only investing in new InsureTech players. Explain how the values of these ratios may turn out to be for such start-ups giving appropriate reasons for the same. (5)

- iii)** Another student actuary says that under IFRS regime the values of these ratios would completely change and hence the valuation of the companies too.

- a)** Discuss how the ratios would change under IFRS regime. (2)

- b)** Comment on the second part of the statement that the valuation of the companies would change under IFRS. (1)
[11]

Q. 5) i) A small recently established insurance company has been writing Motor, Health and Property business and has the following reinsurance arrangements in place:

- Quota Share
- Surplus Treaties
- Excess of Loss Treaties
- Aggregate Excess of Loss Treaties/ CAT Excess of Loss treaties

Explain how the reinsurance arrangements would have been structured to achieve adequate reinsurance cover at the optimal cost. (4)

ii) The company is now considering underwriting crop insurance schemes.

- a)** Outline the characteristics of crop insurance schemes? (2)
- b)** List down the challenges faced by actuaries in crop insurance. (5)
- c)** Discuss the different approaches which could be used for determining various kinds of reserve related to crop insurance? (10)
[21]

Q. 6) You are an actuary working for a large general insurance company writing Motor, Fire and Health business. The company's management is considering acquiring another medium sized general insurance company with similar business. The CEO has asked you to evaluate the proposal. Outline the major inspections you would undertake on the company to be acquired to evaluate the proposal. [10]
